Subject: Front End Technologies

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ASSIGNMENT 1

1. What is Figma? Explain its key features and why it is widely used in modern UI/UX design.
    - Figma is a digital tool used to design how websites and mobile apps look on the screen, such as designing a login page or a home screen. It works fully in a web browser, so a designer can open it on Chrome without installing any software.
    - One important feature is real-time collaboration, where two designers can edit the same screen at the same time. For example, one person can design buttons while another edits text.
    - Figma also supports prototyping, which means a designer can click buttons and move between screens like a real app. Auto layout helps designs adjust automatically, such as a button resizing when text length increases.
    - Designers use components to reuse items like the same button on multiple pages. All work is saved automatically in the cloud, so designs are not lost if the system shuts down.
    - Developers can easily check spacing and sizes, like seeing button width before coding.
    - Teams prefer Figma because design, feedback, and testing happen in one place. This reduces file sharing and saves time during the project.

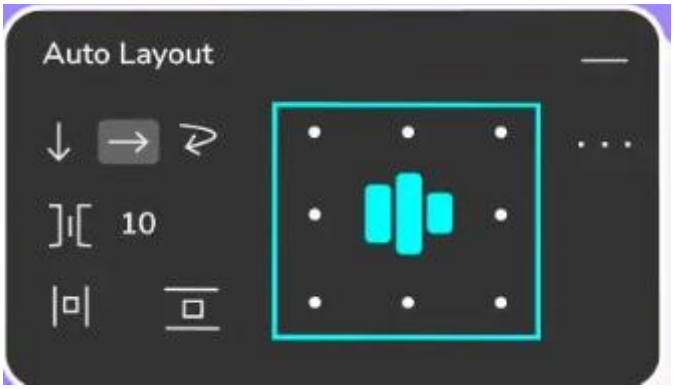
2. Differentiate between Designing and Prototyping in Figma. Explain with one real-world UI example.
    - In Figma, Designing and Prototyping represent two stages of the UI/UX development process. Designing focuses on the visual appearance, while Prototyping focuses on the interactive experience.
    - Designing:
        - Designing is the phase where you build the look and feel of your interface. You spend this time arranging shapes, text, colors, and images on a canvas. In Figma, this involves using the Design Panel to set specific properties like font size, border radius, and layer opacity. It is like creating a "blueprint" what the final product should look like.
    - Prototyping
        - Prototyping is the phase where you add behavior to your static designs. You switch to the Prototype Panel to create "connections" between different frames and components. This allows you to define what happens when a user interacts with an element—such as clicking a button or swiping a card. It simulates the flow of an actual app without requiring any backend code.

- Example:
    - In a food delivery app, designing means creating the home screen with food cards, prices, and buttons.
    - Prototyping means linking the “Order Now” button to the payment screen.

3. What are Components and Styles in Figma? How do they help in maintaining consistency and saving time in large projects?
    - Components are reusable design elements which can be created once and used multiple times anywhere in your whole project.
    - For example: a button can be a component with its shape, color, border.
    - This button can be reused multiple times anywhere in the project. If the attribute of the component is changed then the same attributes change everywhere.
    - In simple words components are basically templates which can be changed and used.
    - Styles work similarly but for specific properties like a specific red color or a bold text size. Instead of remembering a color code, you just select the saved style from a simple list.
    - These features are essential for large projects where multiple designers and developers work together. This prevents errors and creates consistency across your application.
    - It also increases user experience as the user are more likely to remember what a button does based on its look, as the same button is show everywhere.

4. Explain Auto-Layout in Figma. How does it help in creating responsive designs for different screen sizes?
    - Auto Layout is a tool in Figma that lets you create frames that grow or shrink. It automatically moves elements when you change the content inside them or resize the whole frame.
    - For example, if you type a longer word inside a button, the button border stretches with it. You can use it to set specific spacing between items, so they stay perfectly always aligned.
    - This is very helpful for making lists where adding a new item moves everything else down automatically.
    - It helps with different screen sizes because you can set elements to fill the available horizontal space. This means a navigation bar can look correct on both a thin phone and a wide desktop.
    - Instead of moving every icon by hand, Auto Layout calculates the gaps for you based on your rules. It removes the need for manual adjustments every time a change is made to the design.

- Modern designers rely on it to ensure their layouts do not break when the screen size changes. Using this feature makes your design behave more like a real website would in a web browser.



5. Describe the complete developer handoff process in Figma. How are designs shared, assets exported, and code snippets generated for developers?
    - Developer handoff is the stage when the design is passed to the development teams which writes the code for the website or application.
    - Developers can access the file in the Dev mode where they cannot edit the file but can access its elements like buttons, frames, etc.
    - When the developer accesses an element, Figma presents entire details about that element like its dimensions, background color, border radius, shadow, etc.
    - Figma will also automatically generate code snippets. This helps the developer write code faster without the need of manually checking element attributes.
    - Elements such as images or graphics are marked as exportable so that the developer can download them in various formats as required.
    - Designers can also share notes or comments with developers to explain user requirements or how a component is supposed to work.
    - In general, the handoff processing is about moving from designing to development.